



Fighting Flacherie, Grasserie, and Pebrine: Detection Techniques in Sericulture

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Abstract

Silkworm diseases such as flacherie, grasserie and pebrine significantly impact the productivity and sustainability of sericulture. Early and accurate detection techniques are vital for disease management and prevention of large-scale crop losses. This review summarizes conventional, molecular and modern diagnostic approaches for identifying these diseases, while highlighting their advantages, limitations and applicability in sericulture practices. The paper also discusses future prospects of integrating advanced molecular tools and farmer-friendly kits to strengthen disease management programs in sericulture.

Keywords: Flacherie, Grasserie, Pebrine, Sericulture, Disease Detection, *Bombyx mori* and Molecular Diagnosis

Introduction :

Sericulture, the art and science of rearing silkworms (*Bombyx mori* L.) for silk production, is one of the most important agro-based industries that provides sustainable livelihoods to millions of rural families across Asia, particularly in India and China (Kumar and Rahmathulla, 2015). Despite being a profitable sector, silkworm rearing is highly susceptible to microbial diseases, which cause severe economic losses (Nataraju *et al.*, 1999). Among the diseases affecting silkworms, flacherie, grasserie and pebrine are considered the most devastating. Flacherie is a complex disease often caused by bacteria or viruses; grasserie is a viral disease caused by *Bombyx mori* nuclear polyhedrosis virus (BmNPV) and pebrine is a protozoan disease caused by *Nosema bombycis* (Rahmathulla, 2012). These diseases not only reduce cocoon yield but also affect silk quality and overall industry sustainability. The development of reliable detection methods is therefore critical for early disease diagnosis and implementation of management strategies (Vijayan *et al.*, 2011). This review provides a comprehensive account of detection techniques, ranging from traditional microscopic methods to advanced molecular and immunological approaches, along with their practical relevance in sericulture.

Detection Techniques

1.Flacherie Detection

Flacherie is one of the most complex and variable diseases of silkworms. It may be caused by bacterial agents such as *Serratia marcescens*, *Streptococcus* and *Bacillus* species, or by viral infections. Environmental stress, poor hygiene and improper feeding practices can

predispose larvae to the disease (Kumar and Rahmathulla, 2015). Early detection is critical since the disease spreads rapidly under unfavourable rearing conditions.

Traditional detection relies on microscopic observation of larval tissues, especially the midgut, which often shows signs of liquefaction. Culture methods help in isolating and identifying pathogenic bacteria; however, these are time-consuming and require well-equipped laboratories (Rahmathulla, 2012). Recent advances have introduced molecular detection approaches such as Polymerase Chain Reaction (PCR) to identify bacterial and viral pathogens with high specificity (Mori and Kawase, 2008). Additionally, immunodiagnostic assays like ELISA have been developed to detect bacterial toxins or viral antigens (Vijayan *et al.*, 2011). Biochemical techniques such as protein profiling and electrophoresis are also being explored for rapid differentiation of strains responsible for flacherie outbreaks (Kumar and Rahmathulla, 2015).

2. Grasserie Detection

Grasserie is a devastating viral disease caused by BmNPV. The disease is characterized by swelling of larvae due to accumulation of polyhedral occlusion bodies in tissues. Affected silkworms exhibit fragile skin and milky white haemolymph. This disease often spreads quickly in rearing houses, especially under humid and unhygienic conditions (Rahmathulla, 2012).

Detection techniques have evolved from conventional microscopic examination to modern molecular diagnostics. The earliest and simplest method is light microscopy, where polyhedral occlusion bodies can be visualized in haemolymph smears (Nataraju *et al.*, 1999). Histopathological studies of midgut and fat body tissues reveal structural changes induced by viral infection (Mori and Kawase, 2008). Serological assays such as ELISA allow detection of viral proteins with reasonable sensitivity (Vijayan *et al.*, 2011). Molecular methods including PCR, quantitative real-time PCR (qPCR), and loop-mediated isothermal amplification (LAMP) provide rapid and accurate detection of viral DNA (Mori and Kawase, 2008). These tools are increasingly being adapted into field-level diagnostic kits that can be used by farmers and extension workers.

3. Pebrine Detection

Pebrine is a chronic protozoan disease caused by *Nosema bombycis*, a microsporidian parasite. It is considered the most dangerous silkworm disease as it is vertically transmitted from infected moths to their offspring through eggs (transovarial transmission) (Zhou and Nagata, 2001). This makes early detection critical to prevent outbreaks.

Traditional detection involves microscopic examination of silkworm eggs, larvae, pupae, or adult moths for the presence of *Nosema* spores. This technique, though simple, requires expertise and may not detect low-level infections (Rahmathulla, 2012). The 'Ostigmatism test,' a mandatory procedure in seed production centers, ensures that infected eggs are discarded before distribution (Nataraju *et al.*, 1999). Advances in molecular biology have enabled detection of *Nosema* DNA using PCR and quantitative PCR, providing high sensitivity and specificity.

(Zhou and Nagata, 2001). Fluorescence in situ hybridization (FISH) is another promising technique that allows direct visualization of the parasite within host tissues (Mori & Kawase, 2008). Recently, lateral flow immunoassays and portable PCR devices have been tested for their applicability in field conditions, offering hope for farmer-friendly diagnosis of pebrine (Kumar and Rahmathulla, 2015).

Table 1. Molecular techniques used to diagnose silkworm diseases and identify pathogens

SL. No	Assay	Diagnosed pathogen	Host	Disease	Target gene/protein
1	PCR	<i>Nosema assamensis</i>	<i>Antheraea assamensis</i>	Pebrine	SSU rRNA
2	PCR	<i>Nosema pernyi</i>	<i>A. pernyi</i>	Pebrine	SSU rRNA
3	PCR	<i>Nucleopolyhedrovirus</i>	<i>A. proylei</i>	Tiger band	Anpr53
4	PCR	<i>Nucleopolyhedrovirus</i>	<i>B. mori</i>	Grasserie	Polyhedrin
5	PCR	<i>Nucleopolyhedrovirus</i>	<i>B. mori</i>	Grasserie	Polyhedrin
6	PCR	<i>Nucleopolyhedrovirus</i>	<i>B. mori</i>	Grasserie	Polyhedrin
7	PCR	<i>Nucleopolyhedrovirus</i>	<i>B. mori</i>	Grasserie	Polyhedrin
8	PCR	<i>Bacillus</i> sp., <i>Pseudomonas</i> sp.	<i>A. assamensis</i>	Bacterial flacherie	16S rRNA
9	PCR	<i>Nucleopolyhedrovirus</i>	<i>B. mori</i>	Grasserie	ie- 1
10	PCR	<i>Iflavirus</i>	<i>B. mori</i>	BmIFV	RNA dependent RNA polymerase
11	PCR	<i>Densovirus</i>	<i>B. mori</i>	Flacherie	VD1 (ORF- 2)
12	PCR	<i>Bombyx mori</i> <i>Bidensovirus</i>	<i>B. mori</i>	Flacherie	DNV2, VD1, VD2
13	Multiplex PCR	<i>Nosema bombycis</i> <i>Nucleopolyhedrovirus</i> , <i>Bidensovirus</i>	<i>B. mori</i>	Pebrine, grasserie, flacherie	SSU, rRNA, Polyhedrin, DNV2 & DNV3
14	Multiplex PCR	<i>N. bombycis</i> <i>Nucleopolyhedrovirus</i> , <i>Bidensovirus</i>	<i>B. mori</i>	Pebrine, grasserie, flacherie	SSU rRNA, DNA polymerase gene and a putative structural protein gene
15	RT- PCR	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	β- tubulin
16	RT PCR	<i>Iflavirus</i>	<i>A. mylitta</i>	AmIV	RdRp
17	RT PCR	<i>Iflavirus</i>	<i>A. pernyi</i>	ApIV	Helicase, RdRp
18	qPCR	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	Small- subunit rRNA gene
19	qPCR	<i>N. bombycis</i> , <i>N. mylitta</i> , <i>N. assamensis</i>	<i>B. mori</i> , <i>A. mylitta</i> , <i>A.</i>	Pebrine	EB1, 16S rRNA, β- tubulin, PTP1, PTP2.

		<i>N. ricini</i>	<i>assamensis,</i> <i>Samia ricini</i>		PTP3, SWP5 and MetAP2
20	qPCR	<i>Iflavirus</i>	<i>A. pernyi</i>	<u>ApIV</u>	<u>RdRp</u>
21	qPCR	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	β - tubulin
22	qPCR	<i>Nucleopolyhedrovirus</i>	<i>A. mylitta</i>	Grasserie	Lef- 1 and Bm actin
23	LAMP	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	SSU rRNA
24	LAMP	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	SSU rRNA
25	LAMP- LFD	<i>Bombyx mori</i> <i>bidensovirus</i>	<i>B. mori</i>	Flacherie	<u>vp</u> gene
26	PCR, LAMP	<i>Nucleopolyhedrovirus</i>	<i>B. mori</i>	Grasserie	pe38
27	LAMP- LFD	<i>Nucleopolyhedrovirus</i>	<i>B. mori</i>	Grasserie	gp41
28	LFA	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	LSU rRNA
29	ELISA	<i>Nucleopolyhedrovirus</i>	<i>B. mori</i>	Grasserie	Whole virus
30	Slide agglutination	<i>Nosema mylitta</i>	<i>A. mylitta</i>	Pebrine	Purified <i>N. mylitta</i> spores
31	Western blot	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	Endospore protein EOB13320
32	Western blot	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	SWP26
33	Western blot	<i>Nucleopolyhedrovirus</i>	<i>B. mori</i>	Grasserie	gp37
34	Western blot	<u>Cypovirus</u>	<i>A. mylitta</i>	<u>Virosis</u>	p65
35	Western blot	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	NbHSWP11
36	Western blot	<i>Nucleopolyhedrovirus</i>	<i>A. pernyi</i>	Grasserie	<u>PolH</u>
37	ELISA	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	SWP5
38	ELISA	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	NbPTP6
39	ELISA	<i>N. bombycis</i>	<i>B. mori</i>	Pebrine	SWP12

Comparative Analysis of Detection Methods

When comparing detection techniques across these three diseases, it becomes evident that traditional microscopic methods are inexpensive and field-friendly but lack sensitivity (Rahmathulla, 2012). Molecular techniques such as PCR, qPCR and LAMP are highly sensitive and specific but require laboratory infrastructure and trained personnel (Mori and Kawase, 2008). Immunological assays like ELISA bridge the gap between traditional and molecular methods by offering reasonably fast and accurate results with minimal technical requirements (Vijayan *et al.*, 2011). Future research should focus on integrating these techniques into portable diagnostic kits to enable real-time disease monitoring at the farmer’s level.

Future prospects

With the advancement of molecular biology and biotechnology, there is a growing opportunity to develop cost-effective, rapid and reliable diagnostic tools for silkworm diseases. CRISPR-based detection systems, nanotechnology-based biosensors, and mobile-phone integrated diagnostic kits are emerging frontiers that could revolutionize disease management in sericulture (Mori and Kawase, 2008). Moreover, establishing centralized diagnostic laboratories and strengthening extension services will ensure that farmers benefit from these innovations. The combination of traditional practices with modern tools is essential for sustaining the sericulture industry against recurring disease threats (Kumar and Rahmathulla, 2015).

Conclusion

Silkworm diseases such as flacherie, grasserie, and pebrine continue to pose serious challenges to sericulture. Detection is the first and most important step in managing these diseases effectively (Rahmathulla, 2012). While traditional methods remain relevant, molecular and immunological approaches have enhanced the accuracy and reliability of diagnosis (Mori and Kawase, 2008). The integration of innovative diagnostic technologies with practical, farmer-level tools will be crucial in reducing economic losses, ensuring disease-free silkworm seed production, and improving the overall productivity of the sericulture industry (Vijayan *et al.*, 2011).

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